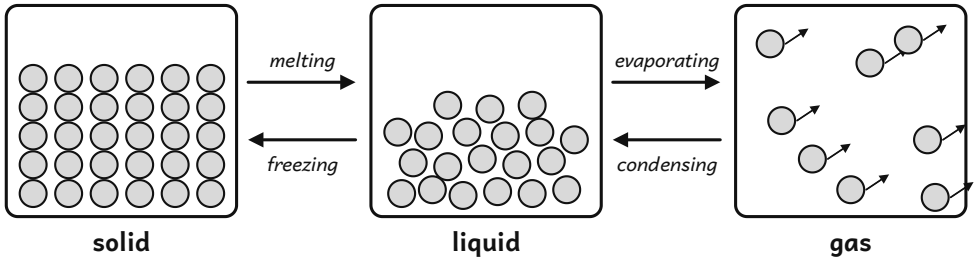
PHYSICAL AND CHEMICAL SCIENCE

States of Matter

Solids, liquids and gases, melting and freezing, and water turning to steam and back



In this pack

- **Learn it:** Solids, liquids and gases ★☆☆
- **Fact check:** Solid, liquid or gas? ★☆☆
- **Think it through:** What will happen? ★★☆
- **Investigate:** Where does ice melt fastest? ★★☆
- **Real-world case study:** The ice cream seller ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science: Stage 2 Unit 3 Changing materials (3.4–3.5 Heating and cooling).

This pack belongs to: _____ **Class:** _____

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LEARN IT · C1 · SOLIDS, LIQUIDS, GASES

Solids, liquids and gases

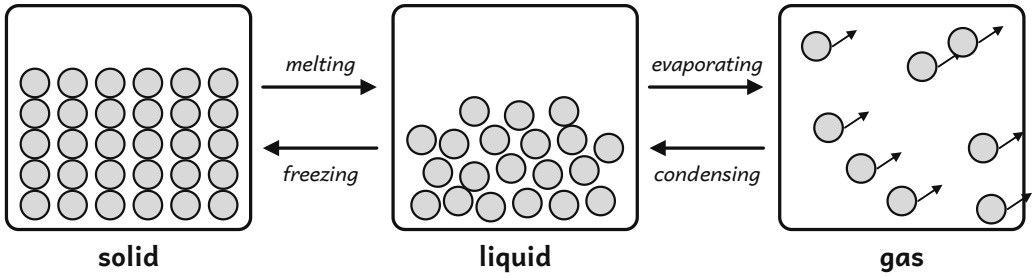
★☆☆ Grade 2

Name: **Class:** **Date:**

Learning goals

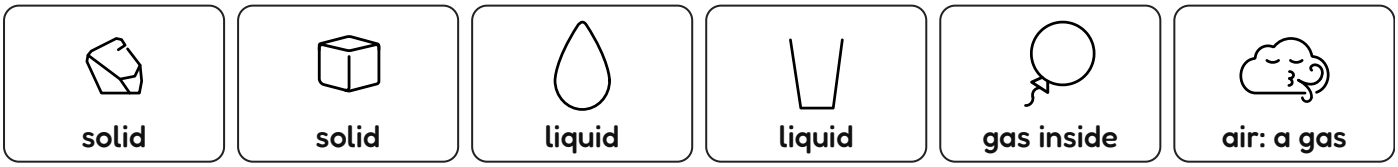
- Sort materials into solids, liquids and gases.
- Explain that matter has mass and takes up space.
- Describe melting, freezing and how water turns to steam and back.

Everything around you is made of **matter**: your desk, the water in your bottle and even the air. Matter has **mass** (it has weight) and takes up **space**. Matter can be a **solid**, a **liquid** or a **gas**.



Solid, liquid and gas.

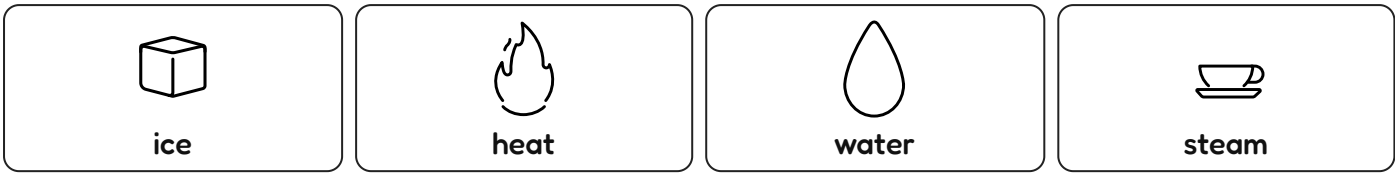
	Solid	Liquid	Gas
shape	keeps its shape	takes the shape of its container	spreads out to fill the space
can you pour it?	no	yes	it floats and spreads
examples	stone, ice, wood, spoon	water, milk, oil, juice	air, steam



Heating and cooling change things

- **Melting**: when a solid is heated it can turn into a liquid. Ice melts into water; chocolate melts in your hand.
- **Freezing**: when a liquid is cooled it can turn into a solid. Water in the freezer becomes ice.
- Melting is the **reverse** (opposite) of freezing.

- When water is heated in a kettle it **boils** and turns into **steam** (a gas). When steam touches a cold lid, it turns back into **water drops**.



Stay safe

Steam and boiling water can burn you. Always ask an adult to help with heating.

Bangladesh connection

In winter in the north of Bangladesh it can be very cold, but water does not freeze outside. It only freezes in a fridge freezer, where it is below 0 °C.

Key words

solid keeps its shape
liquid can be poured
gas spreads out to fill a space
melt a solid turns into a liquid
freeze a liquid turns into a solid
steam water as a gas

FACT CHECK · C1 · SOLIDS, LIQUIDS, GASES

Solid, liquid or gas?

★☆☆ Grade 2

Name: _____ **Class:** _____ **Date:** _____

Learning goals

- Sort materials.
- Name changes.

1 Write solid, liquid or gas. [5]







2 Sort them. [6]

SORT THESE

brick ice oil air juice steam

Solid	Liquid	Gas

3 Fill in the gaps. [3]

WORD BANK

freezes sand melts steam

Ice _____ into water when it is warm. Water _____ into ice in a freezer. Boiling water turns into _____.

What will happen?

★★☆ Grade 2

Name: Class: Date:

Learning goals

- Predict changes when heating and cooling.

1 **What happens to chocolate on a hot day? Can you change it back?** [2]

2 **Rina puts a lid on a pot of boiling water. Later the lid is wet inside. Why?** [2]

3 **Draw what happens to an ice cube left in the sun.** [2]

start	after 1 hour

INVESTIGATE · C1 · SOLIDS, LIQUIDS, GASES

Where does ice melt fastest?

★★☆ Grade 2

Name: Class: Date:

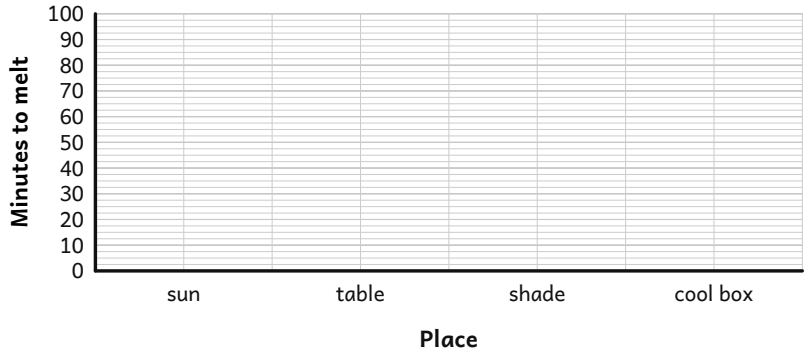
Learning goals

- Do a fair test.
- Time and record.

Class 2A put the same size ice cube in four places and timed how long each took to melt.

1 Draw a bar chart. [3]

Place	in the sun	on a table	in the shade	in a cool box
Minutes to melt	10	25	30	90



2 Where did the ice melt fastest? Why? [2]

3 Why did they use the same size ice cubes? [1]

REAL-WORLD CASE STUDY · C1 · SOLIDS, LIQUIDS, GASES

The ice cream seller

★★★ Grade 2

Name: **Class:** **Date:**

Learning goals

- Use melting and freezing in real life.

Science | Food | Maths

> On hot afternoons, an ice cream seller pushes his cart near the school. Inside the cart is a box with thick walls and lots of **ice**. The ice keeps the ice cream cold so it does not **melt**.
>
> In the morning he buys ice from the ice factory, where water is **frozen** in big machines. By evening most of the ice has melted into water.
>
> A story about an ice cream cart.

1 **Why does the seller need ice in his cart?** [2]

2 **What happens to the ice by evening? Is it a solid or a liquid now?** [2]

3 **The seller sells 6 ice creams every hour for 4 hours. How many does he sell?** [1]

Show what you know

★★★
Grade 2

Name: Class: Date:

Learning goals

- Use matter words.

1 Find the words.

G	K	T	S	S	I	N	E
A	R	R	D	T	R	C	W
S	O	L	I	D	E	P	E
O	N	D	H	J	R	A	U
L	I	Q	U	I	D	V	M
N	D	M	E	L	T	L	O
K	N	L	H	T	A	N	O
E	N	D	C	A	A	K	G

gas ice liquid melt solid steam

2 Which is a liquid? [1]

◯ a stone ◯ milk

◯ air

3 Water turns into ice when it... [1]

◯ melts ◯ freezes

◯ boils

4 Challenge. Is air matter? How do you know? [2]

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
sort solids, liquids and gases	○	○	○
say that matter takes up space	○	○	○
explain melting and freezing	○	○	○
explain steam turning back to water	○	○	○
test where ice melts fastest	○	○	○